

Beiyan Liu

AI Full-Stack Engineer | Applied AI Research | Knowledge Systems

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barytes.github.io/knowledge-base | barytes.substack.com

Profile

- **Product architecture mindset:** Able to define problem value, user scenarios, context boundaries, and success criteria, then design AI product architecture and AI Agent architecture.
- **Algorithm design ability:** Able to model real-world problems mathematically and design operations research and game-theoretic algorithms to solve them.

Selected AI Product Work

gogo – Local Knowledge Workbench for AI-Assisted Research

github.com/Barytes/gogo

- Product function: built an **llm-wiki** desktop knowledge-base app for local document viewing and AI-assisted research workflows.
- Architecture judgment: focused on **llm-wiki**, combining document viewing and Agent chat to reduce setup and switching cost while staying beginner-friendly out of the box.
- Technical implementation: supports Agent chat, sessions, model setup, permissions, and skills; uses Tauri + FastAPI + Pi RPC for streaming chat and local files.

oh-share-it – Shared Research Knowledge Base

github.com/Barytes/oh-share-it

- Product function: built a shared Agent Context layer for research teams, letting members share Agent context by rules so people and Agents can retrieve, reuse, and compound team knowledge.
- Architecture judgment: addressed scattered personal context and Agents' inability to learn from one another by using a federated sharing architecture: members maintain local knowledge bases, while the public layer handles indexing, sync, and routing.
- Technical implementation: supports rule-based file filtering and permission checks; generates L0 / L1 / L2 indexes over resource / memory / skill entries, with CLI and Web UI access.

my-little-chatting-agent – AI Research Assistant Prototype

github.com/Barytes/my-little-chatting-agent

- Product function: built an AI research assistant for web search, page reading, local-note QA, and external information scans.
- Agent implementation: built the tool-calling agent loop from scratch, supporting tool selection, execution, observation feedback, and multi-step reasoning.
- RAG implementation: built a local-note RAG retrieval tool with FAISS and integrated it into the Agent answer flow.

Research

Strategic User Offloading and Service Provider Pricing in Mobile Edge Computing

ICASSP 2026 (CCF-B)

First author

- Modeled strategic interactions among an edge service provider, a network service provider, and users as a two-stage Stackelberg game covering provider pricing and user resource-purchasing decisions.
- Formulated user offloading competition as a generalized Nash equilibrium problem, proved equilibrium existence, and designed a greedy algorithm for the socially optimal user equilibrium.
- Designed an incremental improvement algorithm for service-provider pricing and a branch-and-bound framework for the Stackelberg equilibrium; simulations efficiently approximate the Stackelberg equilibrium.

Efficient and Fair Computing Power Provision for Edge Systems

Computer Networks (CCF-B), under review

First author

- Proposed a cooperative edge computing framework where an edge service provider recruits nearby mobile devices to handle peak demand while fairly incentivizing self-interested contributors.
- Formulated task dispatch, computing-power allocation, and reward distribution as a Nash bargaining problem; decomposed it into social-welfare maximization and optimal reward distribution subproblems.
- Designed an ADMM-based distributed algorithm and a negotiation algorithm; simulations show up to 140% social-welfare improvement and up to 60% maximum-delay reduction over baselines.

Education

Sun Yat-sen University

Guangzhou, China

M.S. in Software Engineering

2023–2026

B.S. in Software Engineering

2018–2022

Writing and Knowledge Systems

Open knowledge base: barytes.github.io/knowledge-base

Blogs: barytes.substack.com | superlinear.academy/u/e062296f